

# **Econometrics I**

## Lecture 13a: Censored Outcomes and the Tobit Model

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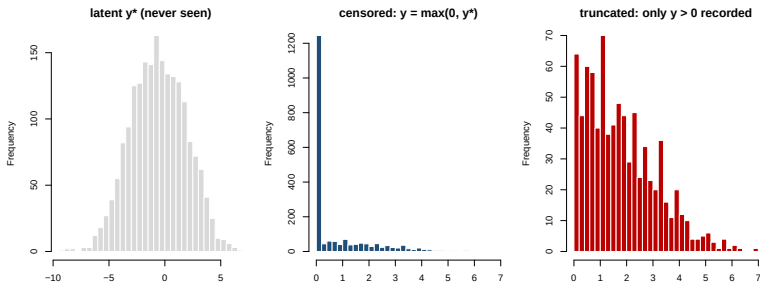
NYU Stern

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## Session 13: outcomes that are not “a number on the real line”

- ▶ Weeks 2–11 modeled  $\mathbb{E}[y \mid x]$  for a continuous  $y$ . Week 12 handled  $y \in \{0, 1\}$  and the case where  $y$  is only *observed* for a selected subsample (Heckman).
- ▶ Today, three more shapes of  $y$  that business data take, each with its own likelihood:
  1. **Censored**: spending is zero for half the sample, or income is top-coded at \$250k. (This deck.)
  2. **Durations**: months until a customer churns, days until a loan defaults, and the clock is still running for many of them when the data end. (Deck b.)
  3. **Counts**: purchases, visits, patents, complaints. Non-negative integers with lots of zeros. (Deck c.)
- ▶ Common thread: OLS on  $y$  answers a question, but usually not the one you asked. Writing down the likelihood forces you to say what  $y$  is, and then Week 12’s MLE machinery does the rest.
- ▶ Readings: Hansen Ch. 27; Wooldridge Ch. 17 (censoring), 22 (durations), 19 (counts).

# Censoring versus truncation



- **Censored**: every unit is in the data, but  $y$  is recorded at a limit when  $y^*$  is past it. You see the pile-up: spending (0), top-coded income, demand at a stockout.
- **Truncated**: units past the limit are *not in the data at all*: only customers who bought appear in the transactions file. Week 12's selection problem with the rule known.
- Censoring is milder: the zeros carry information.

$$y_i^* = x_i' \beta + \varepsilon_i, \quad \varepsilon_i \mid x_i \sim N(0, \sigma^2), \quad y_i = \max(0, y_i^*).$$

- ▶  $y^*$  is the **latent** variable: desired spending, which can be negative (“I would pay to not have this”). We observe it only when it is positive.
- ▶ Two readings of the same equation:
  - **Genuine censoring**:  $y^*$  is real and interesting (true income behind a top-code). The estimand is  $\beta$ .
  - **Corner solution**:  $y^*$  is a device; the economic object is the observed  $y$  (spending is zero because the optimum is a corner). The estimand is  $\partial \mathbb{E}[y \mid x] / \partial x$ .

Same likelihood, different marginal effect. Decide which you are before reporting a number.

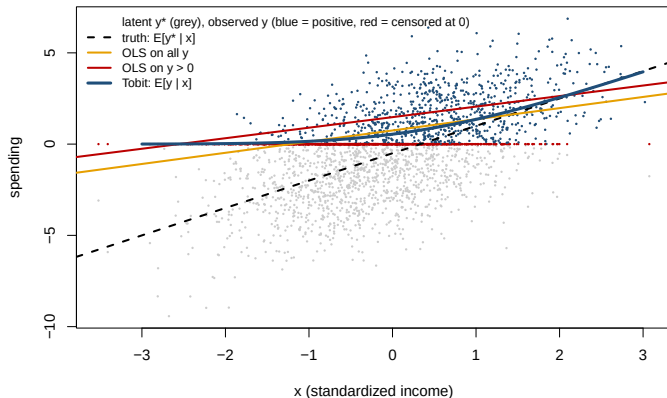
- ▶ Tobin (1958) wrote this down for household durable spending; hence “Tobit.”

## Running example: spending on a product category

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- ▶ 2,000 households.  $x$  is standardized income. Latent spending  $y^* = -0.5 + 1.5x + \varepsilon$ ,  $\varepsilon \sim N(0, 2^2)$ , observed  $y = \max(0, y^*)$ .
- ▶ About 59% of households spend nothing. That is not unusual: most product categories are bought by a minority of households in any given period.
- ▶ Ground truth:  $\beta = 1.5$ ,  $\sigma = 2$ . We will run the regressions people actually run, and then the right one.

## What OLS does



Grey: the latent  $y^*$  nobody sees. Red: households censored at zero. OLS on everyone is pulled flat by the zeros; OLS on buyers only is pulled flat by truncation from below. Both are less than half the truth.

## Why both OLS regressions fail

- **OLS on all  $y$ .** We are fitting  $\mathbb{E}[y \mid x]$ , and

$$\mathbb{E}[y \mid x] = \mathbb{P}(y^* > 0 \mid x) \cdot \mathbb{E}[y^* \mid x, y^* > 0] = \Phi\left(\frac{x'\beta}{\sigma}\right) x'\beta + \sigma \phi\left(\frac{x'\beta}{\sigma}\right),$$

which is *nonlinear* in  $x$  and flatter than  $x'\beta$  wherever censoring bites. A line through it has slope  $\approx \beta \cdot \mathbb{P}(y > 0)$ . Here that is  $1.5 \times 0.41 \approx 0.6$ .

- **OLS on  $y > 0$ .** Conditioning on  $y^* > 0$  means conditioning on  $\varepsilon > -x'\beta$ . Low- $x$  households are in the sample only if they drew a large positive  $\varepsilon$ :

$$\mathbb{E}[y \mid x, y > 0] = x'\beta + \sigma \lambda\left(\frac{x'\beta}{\sigma}\right), \quad \lambda(z) = \frac{\phi(z)}{\Phi(z)}.$$

The omitted term is decreasing in  $x$ : the same inverse Mills ratio as Heckman, with the selection equation known exactly.

- Neither is “biased a little.” Both are wrong by a factor that depends on how much censoring there is.

## The Tobit likelihood: a probit and a regression glued together

Each observation contributes one of two pieces:

$$\ell(\beta, \sigma) = \sum_{i: y_i=0} \ln \underbrace{\left[1 - \Phi\left(\frac{x_i'\beta}{\sigma}\right)\right]}_{\mathbb{P}(y_i^* \leq 0)} + \sum_{i: y_i>0} \ln \underbrace{\left[\frac{1}{\sigma} \phi\left(\frac{y_i - x_i'\beta}{\sigma}\right)\right]}_{\text{normal density at } y_i}$$

- Censored observations enter through a **probability** (we know only that  $y^* \leq 0$ ); uncensored ones through a **density** (we know  $y^*$  exactly). Mixing the two is what makes it a censored model rather than a probit or a regression.
- Truncated regression drops the first sum and divides the density by  $\Phi(x_i'\beta/\sigma)$ : less information.
- Concave in  $(\beta/\sigma, 1/\sigma)$  (Olsen 1978), so Newton converges from anywhere. Only  $\beta/\sigma$  enters the probit part, so a probit on  $\mathbb{1}\{y > 0\}$  estimates  $\beta/\sigma$ : a useful check.

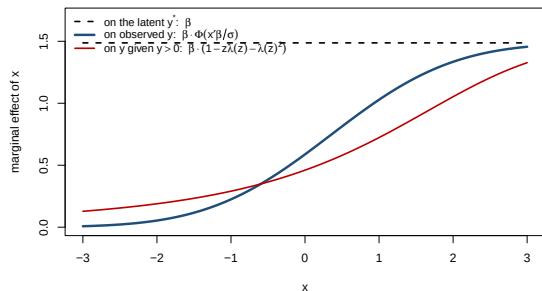


## Estimates on the simulated households

|                            | OLS on $y$ (all) | OLS on $y$ ( $y > 0$ only) | Tobit MLE        | Probit on $\mathbb{1}\{y > 0\}$   | OLS on $y$ |
|----------------------------|------------------|----------------------------|------------------|-----------------------------------|------------|
| Slope on $x$ (truth = 1.5) | 0.612<br>(0.024) | 0.577<br>(0.050)           | 1.487<br>(0.062) | $0.778 = \beta/\sigma$<br>(0.037) | 1<br>(0)   |
| Intercept (truth = -0.5)   | 0.748            | 1.477                      | -0.516           | -0.281                            | -          |
| $\hat{\sigma}$ (truth = 2) | 1.064            | 1.224                      | 1.941            | —                                 | 2          |
| $N$                        | 2000             | 819                        | 2000             | 2000                              | 2          |

- Tobit recovers  $\beta$  and  $\sigma$ . The probit on the zero/positive split recovers  $\beta/\sigma = 0.75$ .
- The last column is what you would get if you could see  $y^*$ : the goal, and Tobit gets there.

## Three marginal effects, and which one you want



- On  $y^*$ :  $\beta$ . Report this if the latent variable is the object (top-coded income).
- On observed  $y$ :  $\beta \cdot \Phi(x'\beta/\sigma)$ , the effect on expected spending including households moved off zero. Report this for corner solutions. It is  $\beta$  times the share of buyers, so it varies with  $x$ .
- On  $y$  given  $y > 0$ :  $\beta [1 - z\lambda(z) - \lambda(z)^2]$ ,  $z = x'\beta/\sigma$ . The intensive margin among current buyers.
- McDonald–Moffitt (1980): the second is the third (intensive) plus the extensive margin.

## Assumptions that do the work

- ▶ **Normality** and **homoskedasticity** of  $\varepsilon$ . Unlike OLS, these are not just for standard errors: the Tobit MLE is *inconsistent* if either fails, because the probability  $\mathbb{P}(y^* \leq 0 \mid x)$  is computed from the assumed distribution.
- ▶ Practical check: does the probit on  $\mathbb{1}\{y > 0\}$  give roughly  $\hat{\beta}_{Tobit}/\hat{\sigma}$ ? If not, the two pieces of the likelihood disagree about the model.
- ▶ **The zeros are the same process as the positives.** Tobit says a household spends zero because its latent demand is low. If zeros come from a *different* decision (never shops this category; the store was out of stock), you want a two-part or hurdle model: a probit for “any?” and a regression for “how much, given any?” with separate coefficients. The count-data deck does the same thing for integers.
- ▶ Heteroskedasticity and non-normality can be handled (CLAD, Powell 1984, needs only a median restriction), at the cost of efficiency and software convenience.

## Variations you will meet

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- ▶ **Top-coding / right-censoring.**  $y = \min(y^*, c)$ . Same likelihood with  $\Phi$  replaced by  $1 - \Phi$  in the censored piece. Wages in public-use files, damage awards, capped insurance claims.
- ▶ **Two-sided censoring.** Scores bounded at 0 and 100; shares in  $[0, 1]$ .
- ▶ **Censoring points that vary by  $i$ .** Each unit has its own limit  $c_i$  (a customer's credit limit; a plant's capacity). Fine, as long as  $c_i$  is known and exogenous.
- ▶ **Interval censoring.** You know  $y^* \in (a_i, b_i]$ : income reported in brackets. The likelihood piece is  $\Phi(b) - \Phi(a)$ . This is also how duration data with grouped time work.
- ▶ **Censoring in time** is the whole subject of the next deck.

Tobit is a left-censored normal regression, which is what `survreg` in the `survival` package (shipped with R) estimates:

```
library(survival)
tob <- survreg(Surv(y, y > 0, type = "left") ~ x, data = d, dist = "gaussian")
coef(tob); tob$scale           # beta and sigma
feglm(I(y > 0) ~ x, d, family = binomial("probit")) # check: should be about beta / sigma
```

```
# statsmodels has no Tobit; the likelihood is eight lines with scipy.optimize
from scipy.stats import norm
def nll(th, y, X):
    b, s = th[:-1], np.exp(th[-1]); xb = X @ b
    return -(np.where(y > 0, norm.logpdf(y, xb, s), norm.logcdf(-xb / s))).sum()
```

► Marginal effects on observed  $y$ : multiply  $\hat{\beta}$  by the average of  $\Phi(x'_i \hat{\beta} / \hat{\sigma})$ .

- ▶ Censored: you see the pile-up at the limit. Truncated: the units past the limit are gone.
- ▶ OLS on censored  $y$  estimates a flattened slope; OLS on the positives has a Heckman-style omitted term. Neither is close.
- ▶ Tobit is a probit for the censored observations glued to a normal regression for the rest. It needs normality and homoskedasticity for *consistency*, not just for inference.
- ▶ Three marginal effects. Say whether your question is about the latent variable or the observed one before you pick.
- ▶ If the zeros are a different decision, use a two-part model. The count deck picks that up.